

Minor Project

Implementing Basic Cloud Security Controls in Microsoft Azure

Scenario

A startup company named XYZ Solutions has recently migrated a small portion of its infrastructure to Microsoft Azure. The company needs a secure environment for storing files and hosting a test server for internal development activities. As a Cloud Security Analyst, your responsibility is to deploy and secure the environment using fundamental Azure security controls while following security best practices.

Project Objectives

- Deploy Azure resources securely.
- Implement storage security controls.
- Configure basic network security.
- Apply access control mechanisms.
- Demonstrate secure cloud deployment practices.
- Document and justify security decisions.

Section A – Theory Assessment

Answer Any Five Questions

1. Explain the Shared Responsibility Model in cloud computing.
2. What is the Principle of Least Privilege (PoLP) and why is it important?
3. Differentiate between Public Blob Storage and Private Blob Storage.
4. What is the purpose of a SAS Token in Azure Storage?
5. Explain the role of Microsoft Sentinel in cloud security.
6. What is the difference between Azure Monitor and Microsoft Sentinel?

Section B – Practical Implementation

Task 1: Secure Storage Deployment

Create and secure an Azure Storage Account.

Requirements

- Create a Storage Account.
- Create a Private Blob Container.
- Upload at least one sample file.
- Disable anonymous/public access.
- Generate a SAS Token with Read permission.
- Access the file using the generated SAS URL.

Demonstration Evidence

- Private container creation.
- Uploaded file.
- SAS Token generation.
- Successful access using SAS URL.
- Public access being denied.

Task 2: Secure Virtual Machine Deployment

Deploy and secure an Azure Virtual Machine.

Requirements

- Create a Linux or Windows Virtual Machine.
- Configure a custom administrator username.
- Allow SSH or RDP access.
- Configure Network Security Group (NSG) rules.
- Restrict unnecessary inbound access.
- Document implemented security controls.

Demonstration Evidence

- VM deployment.
- NSG configuration.
- Successful remote access.
- Security settings applied.

Section C – Security Analysis

Storage Security

- Why was a private container used?
- Why was SAS chosen instead of public access?

VM Security

- Why are NSG rules important?
- What risks are reduced by restricting inbound access?

Recommendations

Provide at least three recommendations to further improve the security posture of the environment.

Final Deliverables

- Answers to Theory Questions
- Architecture Diagram
- Storage Configuration Screenshots
- VM Configuration & Access Screenshots
- NSG Configuration Screenshots
- SAS Access Demonstration Screenshots
- Security Assessment Report (2–3 Pages)

Submission Guidelines

- All screenshots must clearly show the configured resources.
- Screenshots should be properly labelled.
- The report should include observations and security justifications.
- Students must explain the purpose of each implemented security control.